

Xiangjian Zeng

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Research Objective

To research brain physiology with computational modeling and create a functional neuromodulation system with engineering toolsets

Education

Massachusetts Institute of Technology, Cambridge, MA

Ph.D in Electrical Engineering and Computer Science, GPA: 5.00/5.00

Carnegie Mellon University, Pittsburgh, PA

M.S. in Electrical and Computer Engineering, GPA: 4.00/4.00

12/2023

University of California San Diego, La Jolla, CA

B.S. in Electrical and Computer Engineering (Minor in Cognitive Science), GPA: 3.91/4.00

12/2020

Honor: Magna Cum Laude

Research Experience

Lewis Lab, Massachusetts Institute of Technology, Cambridge, MA

09/2024-present

Closed-loop Auditory Modulation of Cerebrospinal Fluid during Sleep

- Design machine-learning-based algorithms to use auditory stimulation to promote cerebrospinal fluid circulation during sleep to enhance waste clearance, suggest new potential intervention for brain aging which does not exist in the field, and present results at conference as first author

Insanally Lab, University of Pittsburgh, Pittsburgh, PA

05/2023-08/2024

Bayesian Ensemble Neural Decoding

- Developed an ensemble decoder in Python that incorporates information from multiple neurons to predict mouse behavior in auditory discrimination tasks, revealed the significance of non-classically-response neurons in task learning, and published the findings in a paper in review

Spiking Recurrent Neural Network Simulation of the Brain

- Designed a spiking Recurrent Neural Network in Python and Julia that follows biological constraints to simulate cerebral mechanisms in mouse auditory discrimination tasks and expand the simulation to discover how a flexible network is attained in long-term training, and first authored a paper in preparation

Chamanzar Lab, Carnegie Mellon University, Pittsburgh, PA

05/2022-04/2023

Neural Probe Waveguide Characterization System

- Created a neural optical probe characterization system based on light scattering, incorporate systems consisting of robotics, computer vision, and signal processing, and published a characterization performance that's previously only obtainable by destructive methods as first author

Lerman Lab, UC San Diego, La Jolla, CA

03/2020-01/2021

Adaptive Neural Signal Noise Cancellation

- Led research and implementation of an adaptive filtering algorithm to eliminate powerline noise contaminating peripheral nerve recordings, prototyped a hardware demo achieving 45dB noise reduction in 60Hz band, and presented a conference paper on project as first author

Peripheral Nerve Magnetoneurography

- Created signal cleaning and processing scripts in Matlab for experiment data on magnetic nervous system recordings by tailoring functions to individual signal, contributing to publication of another paper
- Ran test subject for human trials in neuroscience studies involving electric stimulation

Spitzer Lab, UC San Diego, La Jolla, CA

10/2019-12/2020

Automated Fluorescence Cell Counting

- Developed an automatic cell recognition program for fluorescence brain slices in Matlab utilizing correlation to count and map a variety of brain cells, obtaining an accuracy that's close to expert's manual selection

Neuroelectronics Group, La Jolla, San Diego, CA

07/2019-08/2019

Neural Recording Amplifier PCB Design

- Created and soldered a PCB design using Altium for an amplifier chip to be used in in vivo cell recording, producing a miniature board with size 3cm*1cm and more than two times smaller than previous design

Publications

Paper Publications

- **Zeng, X.**, Reddy, J., Chamanzar, M. Automated Photonic Waveguide Loss Measurement using Out-scattering Light Method. Integrated Systems Design and Technology Conference 2023 book proceedings.
- Toth, J., Sidleck, B., Lombardi, O., Hou, T., Eldo, A., Kerlin, M., **Zeng, X.**, Saeed, D., Agarwal, P., Leonard, D., Andrino, L., Inbar, T., Insanally, M. Dynamic gating of perceptual flexibility by non-classically responsive neurons (manuscript in review).

Conference Publications

- **Zeng, X.**, Levitt, J., Lewis, L.D. Auditory Stimulation During Sleep Modulates CSF Flow via Dual Slow-Wave and High Frequency Pathways. Society for Neuroscience, San Diego, USA, Nov 15-19, 2025.
- **Zeng, X.**, Toth, J., Albana, B. F., Insanally, M. N. Contributions and synaptic basis of diverse cortical neuron response to flexible task performance. Society for Neuroscience, Chicago IL, United States, Oct 5-9, 2024.
- **Zeng, X.**, Yu, G., Albana, B. F., Insanally, M. N. Contributions and synaptic basis of diverse cortical neuron response to flexible task performance. COSYNE, Montreal, Canada, Mar 27-30, 2024.
- **Zeng, X.**, Lerman, I., Bu, Y., Kimball, D. RLS Harmonic Powerline Noise Cancellation in Radial Nerve Recording. North American Neuromodulation Society Virtual Meeting, Orlando FL, United States, Jan 15-16, 2021.
- Bu, Y., Lerman, I., Borna, A., Coleman, T., Rao, R., Huang, M., Shah, V., Kimball, D., Morton, T., **Zeng, X.** Capturing the Magnetic Signature of Small Fiber Neuronal Activity. North American Neuromodulation Society Virtual Meeting, Orlando FL, United States, Jan 15-16, 2021.
- Hou, T., Toth, J., Sidleck, B., Lombardi, O., Eldo, A., Saeed, D., Kerlin, M., **Zeng, X.**, Agarwal, P., Leonard, D., Andrino, L., Inbar, T., Insanally, M. N. Dynamic gating of perceptual flexibility by diverse cortical responses. Society for Neuroscience, 2023.

Fellowship Awards

- **MIT Artificial Intelligence and Wellness Fellowship** 2025

Academic Project

Deep Learning Automatic Fake News Detection

Fall 2022

- Jointly created a BERT based language model to automatically extract relevant evidence and classify misinformation from political statements, achieving a two-way accuracy of 72%

EEG-based Music Recommendation

Fall 2022

- Led to write a K Nearest Neighbor(KNN) prediction system to automatically detect musical preferences from brain EEG recordings of listening subjects and update music recommendation in real-time

Relevant Courses

Computing: CMU - Intro to Deep Learning, Machine Learning for Signal Processing

Neuroscience: MIT - Functional Magnetic Resonance Imaging: Data Acquisition and Analysis, CMU - Neuroengineering Laboratory, Advanced System Neuroscience, UCSD - Cognitive Neuroscience

Electrical Engineering: UCSD - Linear Electronic Systems, Introduction to Digital Design, Components and Circuits Laboratory

MEMS: CMU - Micro and Nano Systems Fabrication, Nano-Bio-Photonics

Technical Skills

Programming Languages: Verilog, C/C++, Matlab, Python, R, Mathematica

Circuit/System Software: Altium, Simulink

Mathematical Packages: Matlab, Mathematica

CAD Software: SolidWorks, Autodesk, Blender

References

Professor Laura D. Lewis, Electrical Engineering and Computer Science, Massachusetts Institute of Technology, ldlewis@mit.edu

Professor Michele Insanally, Otolaryngology & Neurobiology, University of Pittsburgh
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Professor Badr F. Albanna, Neuroscience, University of Pittsburgh School of Medicine
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